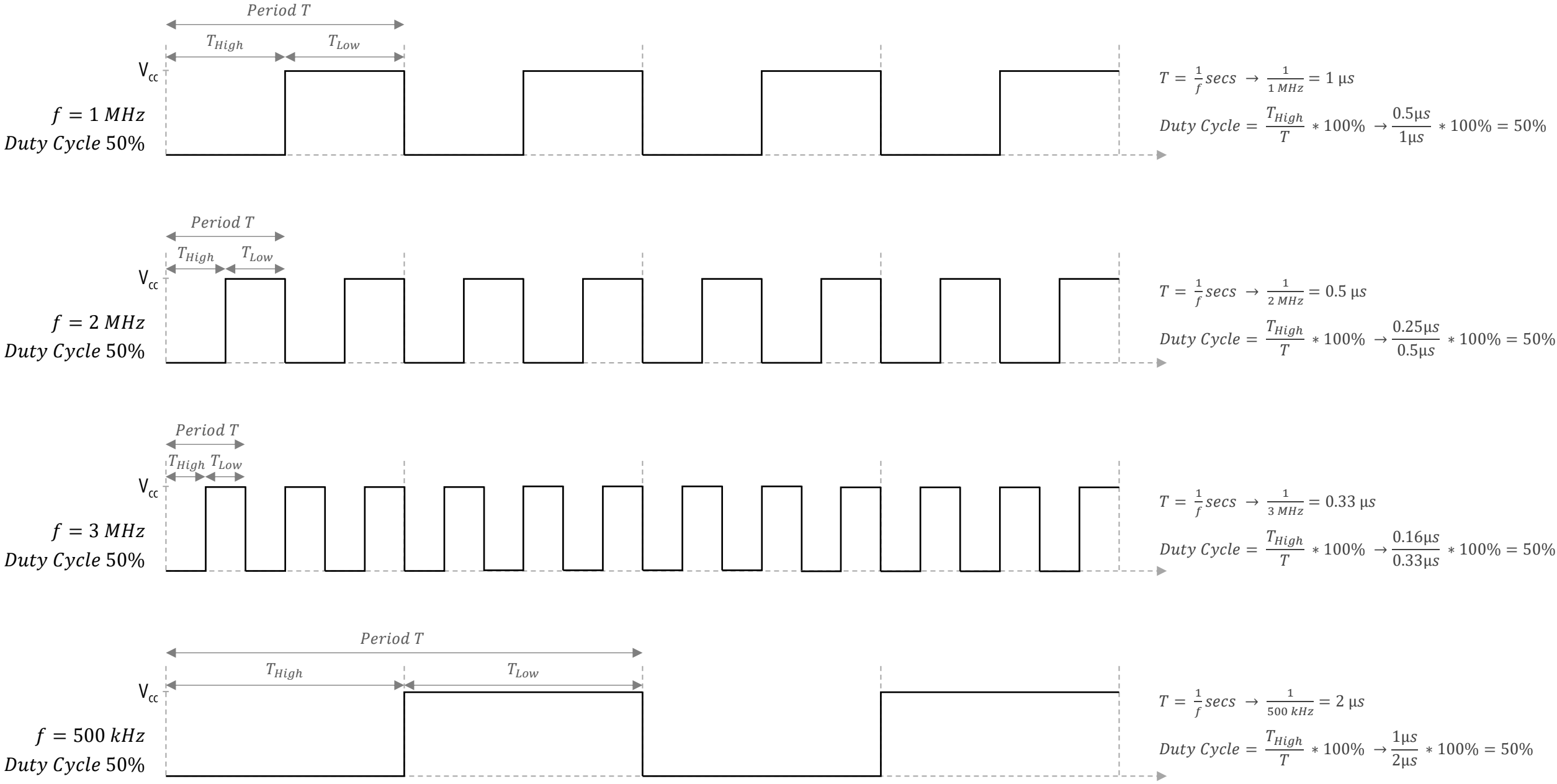
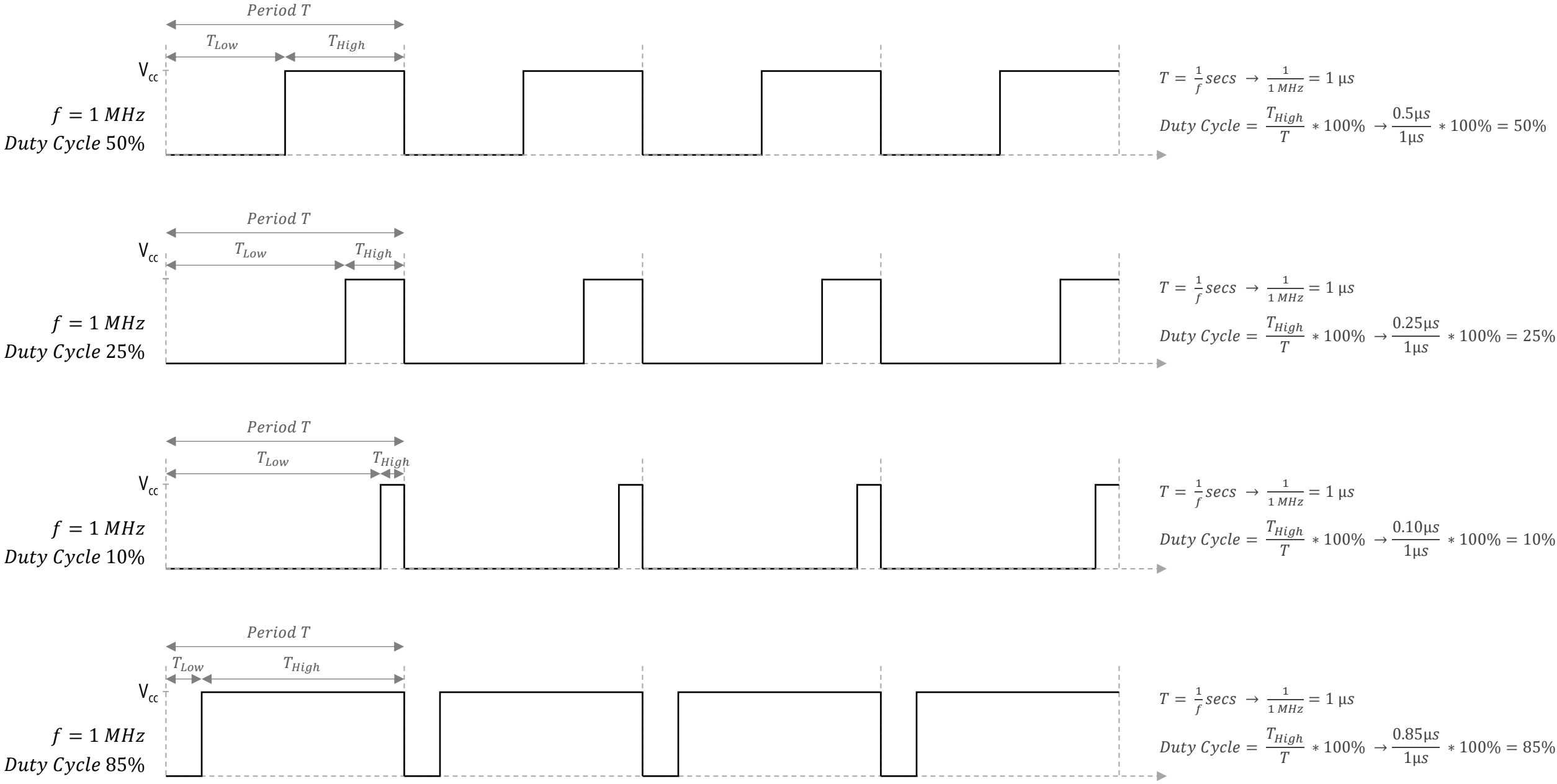


SAME DUTY CYCLE (50%), DIFFERENT FREQUENCIES

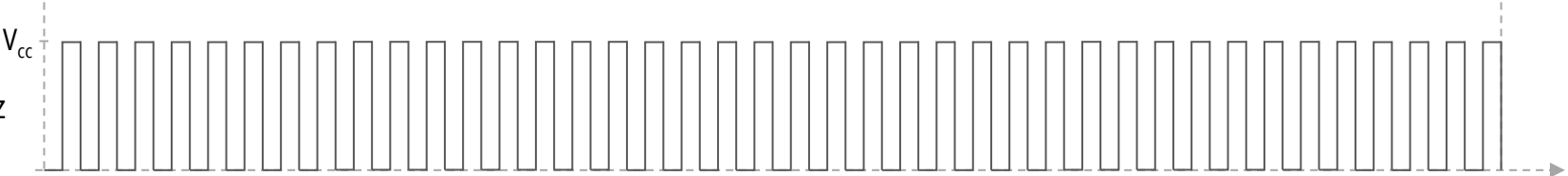


DIFFERENT DUTY CYCLE, SAME FREQUENCY (1MHZ)

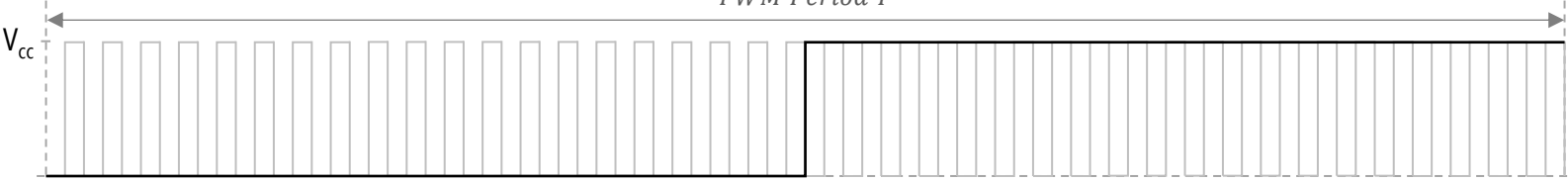


FREQUENCY AND DUTY CYCLE RESOLUTION

Clock frequency = 40 MHz



PWM frequency = 1 MHz
PWM duty cycle = 50%



PWM resolution = 1 bit

2 bits

3 bits

4 bits

5 bits

6 bits

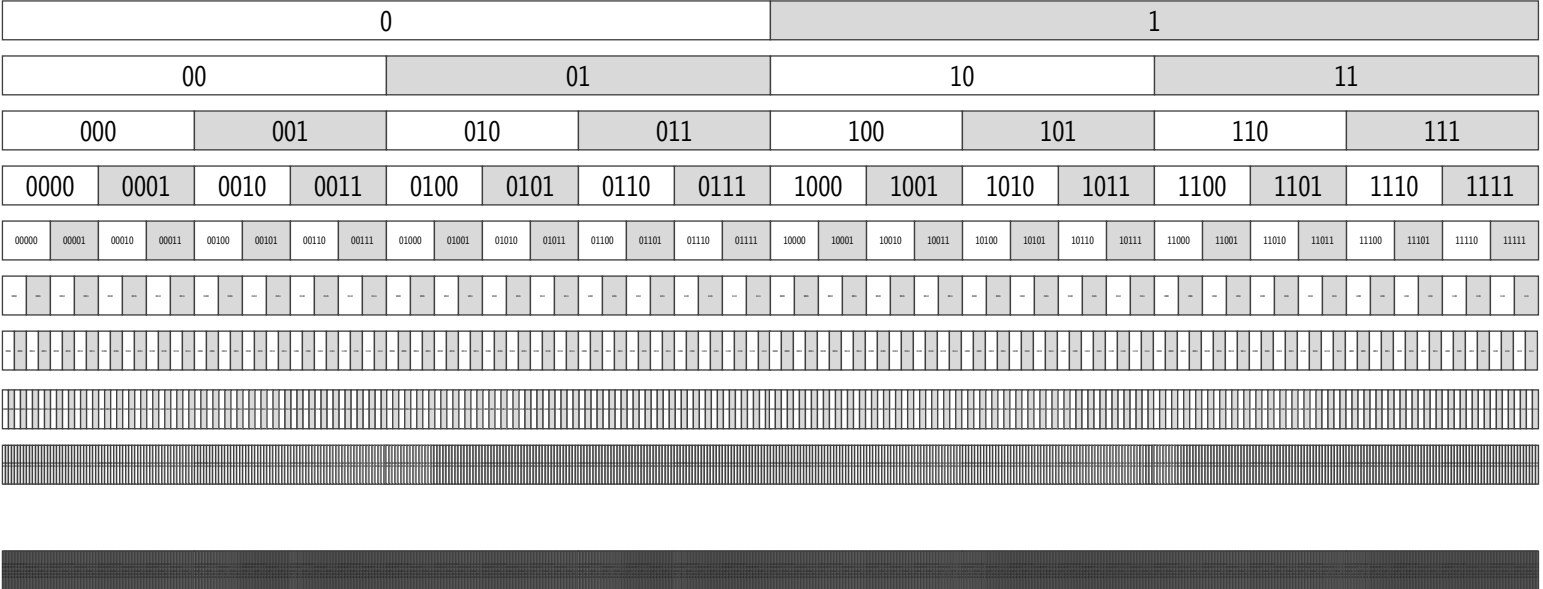
7 bits

8 bits

9 bits

...

16 bits



Bits	Duty Cycle Resolution	Voltage Step Size (V _{cc} = 3.3V)	Minimum Required Clock* with PWM freq=1MHz
1	Fixed at 50%	Fixed at 1.65V	2 MHz
2	25%	825 mV	4 MHz
3	12.5%	412.5 mV	8 MHz
4	6.25%	206.25 mV	16 MHz
5	3.125%	103.13 mV	32 MHz
6	1.5625%	51.56 mV	64 MHz
7	0.78125%	25.78 mV	128 MHz
8	0.39063%	12.89 mV	256 MHz
9	0.19531%	6.45 mV	512 MHz
...			
16 bits	0.001531%	0.0504 mV	65.536 GHz

*Min clock freq = PWM freq * 2^{resolution}